PAPER SUMMARY

AION: Adaptive Inference Optimization Network for High-Throughput, Hallucination-Aware, and Feedback-Driven Large Language Model Serving

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ABSTRACT

Large language model serving systems optimize throughput, accuracy, and adaptation independently. This creates a structural problem: high-throughput serving degrades output quality, hallucination detection runs as a disconnected post-processing step, and adapter fine-tuning requires serving restarts. No deployed inference engine coordinates all three.

This paper presents AION (Adaptive Inference Optimization Network), a unified LLM inference framework coordinating throughput optimization, confidence-aware hallucination detection, and feedback-driven LoRA adapter management through a shared control loop. AION introduces five layers: Request Router, Inference Core, Confidence Estimator, Adapter Orchestrator, and Feedback Controller. Experimental evaluation on a Tesla T4 GPU using OPT-1.3B and OPT-2.7B demonstrates: 40× throughput improvement from batch 1 to 8 (10.2 to 412.9 tokens/second); confidence score separation between flagged and passed outputs (gap 0.2006 at 1.3B, 0.3323 at 2.7B); negligible confidence overhead (–0.27%); and a novel finding that threshold calibration must be domain-sensitive, as larger models exhibit a confidently-wrong failure mode on knowledge-cutoff content.

HIGH-LEVEL ARCHITECTURE

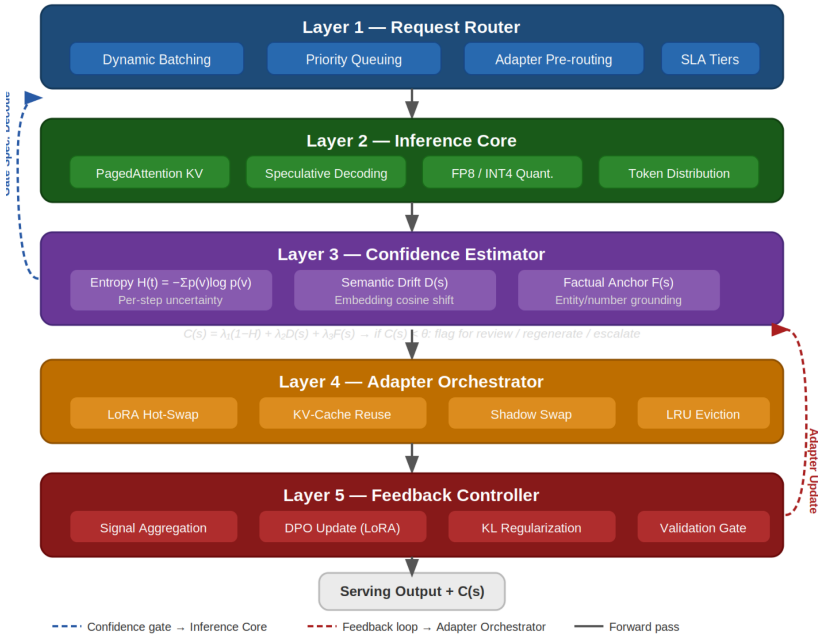


Fig. 1. AION five-layer inference architecture: Request Router, Inference Core, Confidence Estimator, Adapter Orchestrator, and Feedback Controller, operating as a unified inference control loop.

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